Lecture 15:

VLSI Phase-Locked Loop Circuits

Contents

This Topic: VLSI Phase-Locked Loop (PLL) Circuits

- Introduction
- PLL Architecture
- Building Blocks
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Introduction

- A Phase-Locked Loop (PLL) is a TDSP elements that realizes the phase scaling operation
- VLSI PLLs are typically design and analyzed in frequency domain
- VLSI PLLs are widely used VLSI systems for
  - Frequency synthesis
  - De-skew clock
  - Clock recovery
  - Modulation/De-modulation
  - ???

TDSP Model of PLL

- Under the TDSP view, PLL realizes a linear phase scaling operation:

$$\phi_o \rightarrow (N/M)\phi_i$$

  - It can be used to generate(i.e. synthesis) a different clock frequency
- Above scaling is true at low frequency
- At high frequency the output phase losses track (bandwidth limited)

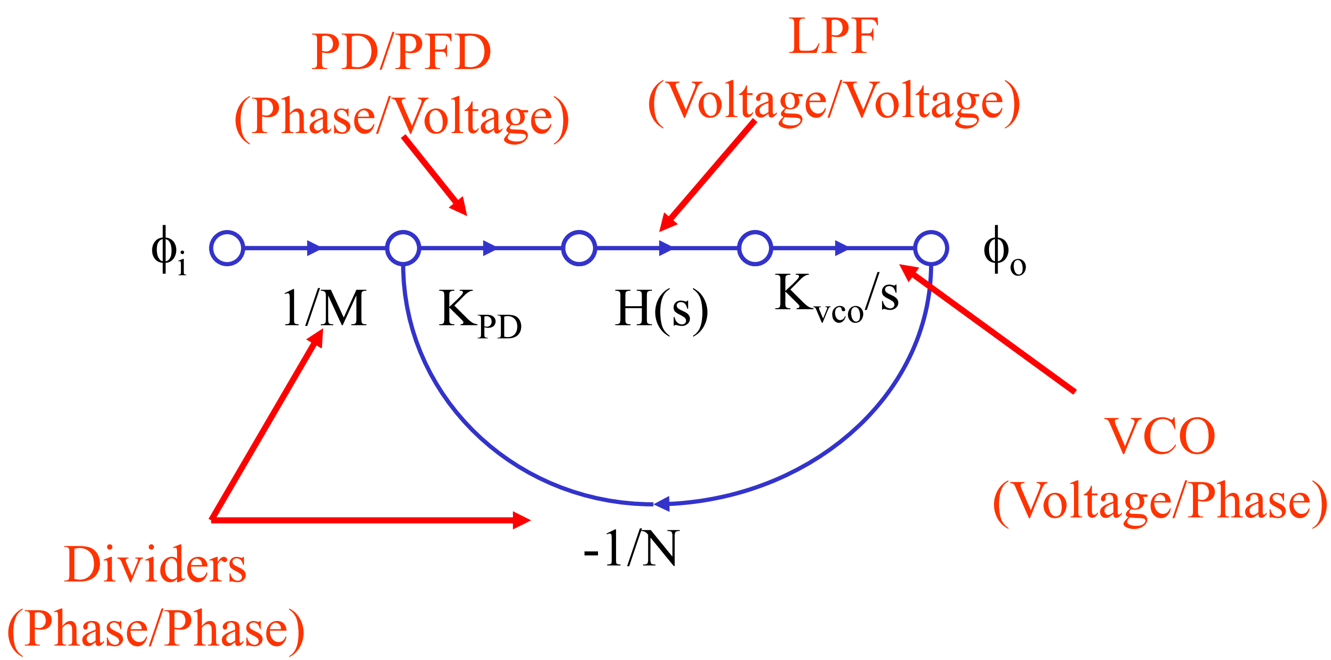
$\phi_i \xrightarrow[N/M]{} \phi_o$ (at low frequency)

$\phi_i \xrightarrow[F(s)]{} \phi_o$ (In general) $F(s)_{|\omega\rightarrow 0} \rightarrow (N/M)$

$F(s)_{|\omega>>\omega o} \rightarrow 0$

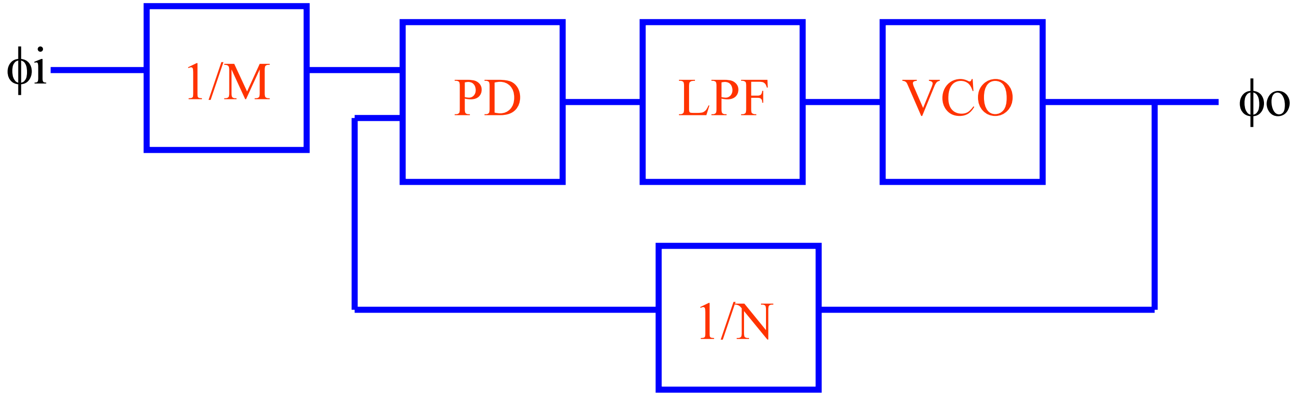
TDSP Model of PLL

- A Linearized S-domain model
  - PLL BW is much lower than clock frequencies (e.g. <1/10 at PD)
  - Small signal model
- A Mixed phase/voltage (or current) domain model
- A s-domain approximation of z-domain system



VLSI PLL Circuit Building Blocks

- Phase Detector/Phase Frequency Detector (PD/PFD)
- Loop Filter (LPF)
- Voltage Control Oscillator (VCO)
- Frequency Dividers (1/N, 1/M)



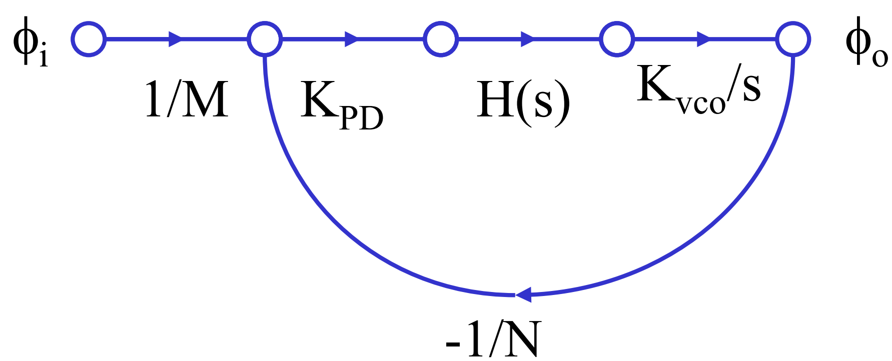
PLL Terminology: Type and Order

- PLL Type: Number of poles at origin in the OPEN loop transfer function

$$\boxed{TF_{Open} \propto H(s)\cdot\frac{K_{VCO}}{s}}$$

- PLL Order: Number of poles in its CLOSE loop transfer function

$$\boxed{\frac{\phi_o}{\phi_i}=\frac{\frac{H(s)K_{PD}K_{VCO}/M}{s}}{1+\frac{H(s)K_{PD}K_{VCO}/N}{s}}}$$



Steady State Phase Tracking Error

- Tracking Error at PD

$$\boxed{\phi_e = \frac{\phi_o}{N} - \frac{\phi_i}{M} = \frac{1}{1 + \dfrac{H(s)K_{PD}K_{VCO}/N}{s}}\phi_i}$$

- Steady State Tracking Error by Final Value Theorem

$$\boxed{\lim_{t\to\infty}\phi_e(t) = \lim_{s\to 0}[s\phi_e(s)] \propto \lim_{s\to 0}[s^{(Type+1)}\phi_i(s)]}$$

Steady State Tracking Error vs. Type

- Three special cases
  - Step position phase input

$$\boxed{\phi_i = C_p H(t) \Rightarrow \phi_i(s) = C_p / s} \quad \Rightarrow \quad \boxed{\lim_{t\to\infty} \phi_e(t) \propto \lim_{s\to 0}[s^{(Type)}]}$$

  - Step velocity phase input

$$\boxed{\phi_i = C_v H(t)t \Rightarrow \phi_i(s) = C_v / s^2} \quad \Rightarrow \quad \boxed{\lim_{t\to\infty} \phi_e(t) \propto \lim_{s\to 0}[s^{(Type-1)}]}$$

  - Step acceleration phase input

$$\boxed{\phi_i = C_a H(t)t^2 \Rightarrow \phi_i(s) = 2C_a / s^3} \quad \Rightarrow \quad \boxed{\lim_{t\to\infty} \phi_e(t) \propto \lim_{s\to 0}[s^{(Type-2)}]}$$

Note: for frequency tracking, type >1 is required

PLL Stability

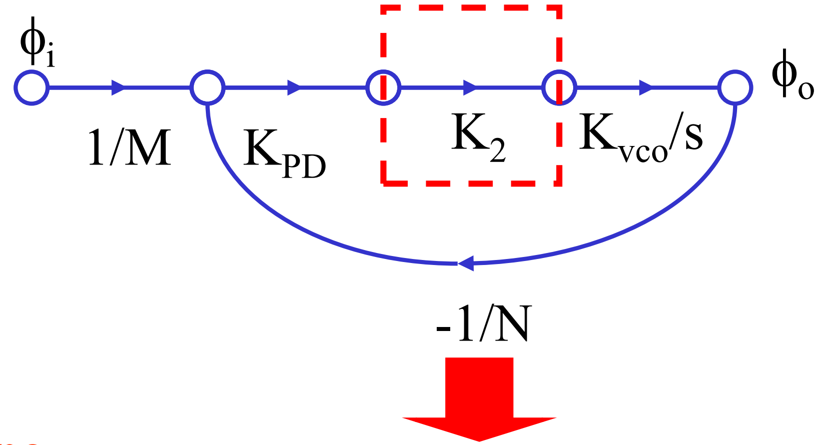
- The pole location of the PLL close loop transfer function determine PLL stability (small signal aspects)
  - Left side of s-plan (stable)
  - Right side of s-plan (unstable)
- For 2nd order PLL stability
  - nature frequency (2d order PLL) <= 1/20 of reference clock frequency (large signal aspects)

First Order PLL Model

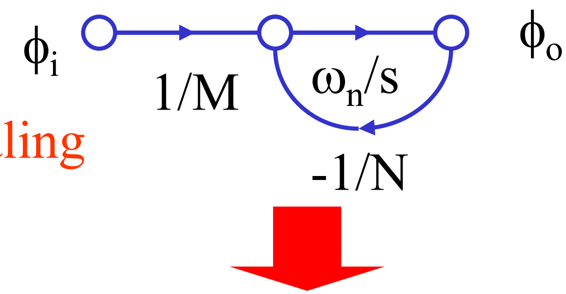
- Parameter: Proportional gain K_P

$$\boxed{H(s) = K_2}$$

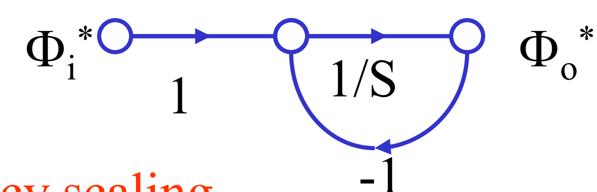
$$\boxed{\frac{\phi_o^*}{\phi_i^*} = \frac{1}{1+S}}$$



Input phase scaling



Output phase scaling



Where

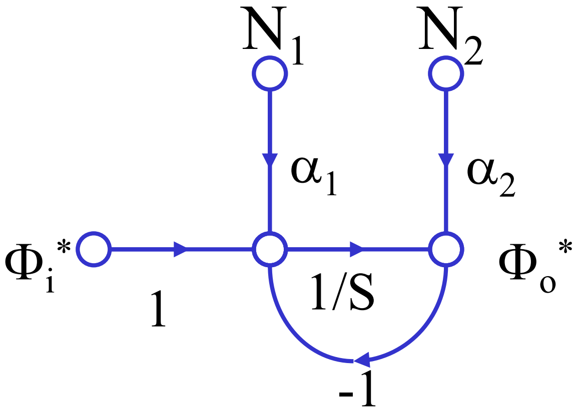
$$\phi_o^* \equiv \phi_o / N$$
$$\phi_i^* \equiv \phi_i / M$$
$$\omega_n \equiv K_{PD} \cdot K_2 \cdot K_{VCO} / N$$
$$S \equiv s / \omega_n$$

Loop response frequency scaling

1^{st} Order PLL Jitter Transfer Function

- Noise N1:

$$\boxed{H(s) = K_2}$$



Input phase scaling

Output phase scaling

Where

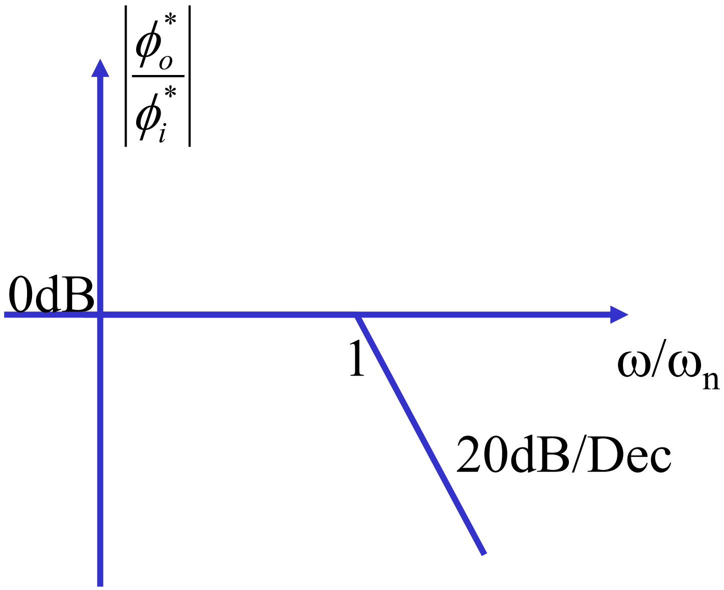
$$\phi_o^* \equiv \phi_o / N$$
$$\phi_i^* \equiv \phi_i / M$$
$$\omega_n \equiv K_{PD} \cdot K_2 \cdot K_{VCO} / N$$
$$S \equiv s / \omega_n$$

Loop response frequency scaling

$$\frac{\phi_o^*}{\phi_i^*} = \frac{1}{1+S}$$

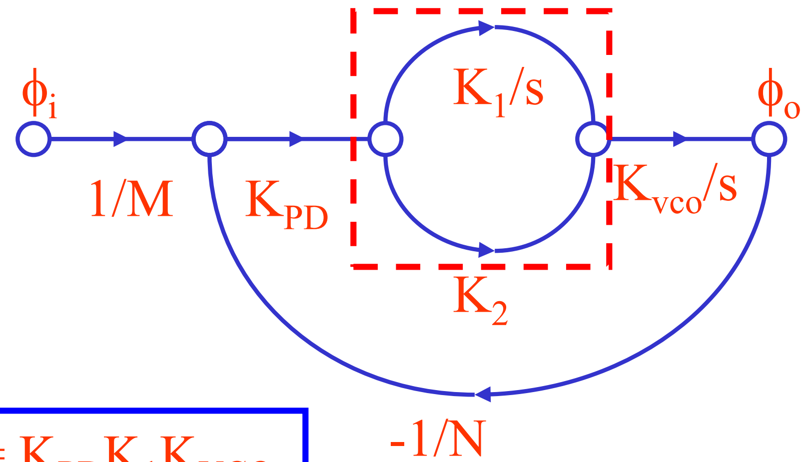
First Order PLL Model

- Frequency Response
  - Lowpass Response
  - -3dB BW = ω_n



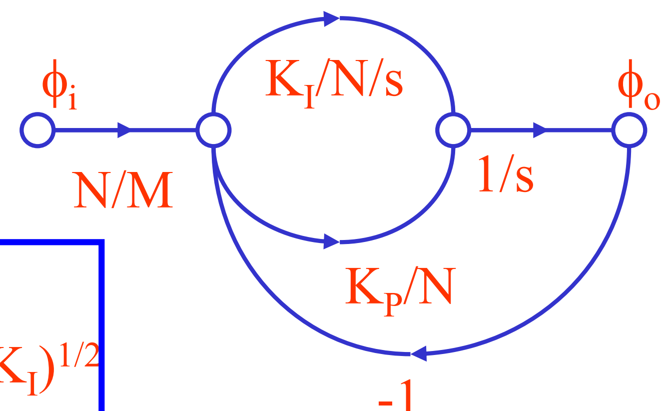
Second Order PLL Model

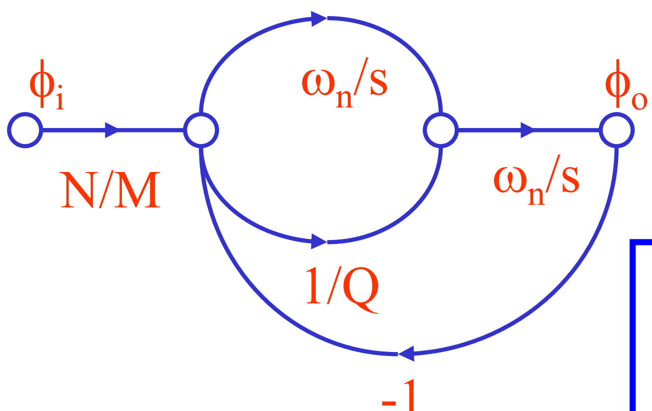
- Parameters:
  - Integral Gain : K_I
  - Proportional Gain: K_P
  - Natural Frequency
  - Q-factor (Damping Factor)



$K_I = K_{PD} K_1 K_{VCO}$

$K_P = K_{PD} K_2 K_{VCO}$



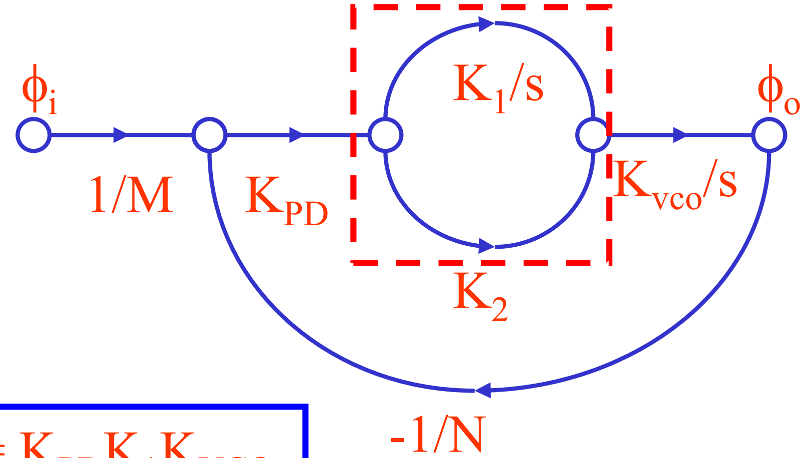


$\omega n = (K_I/N)^{1/2}$

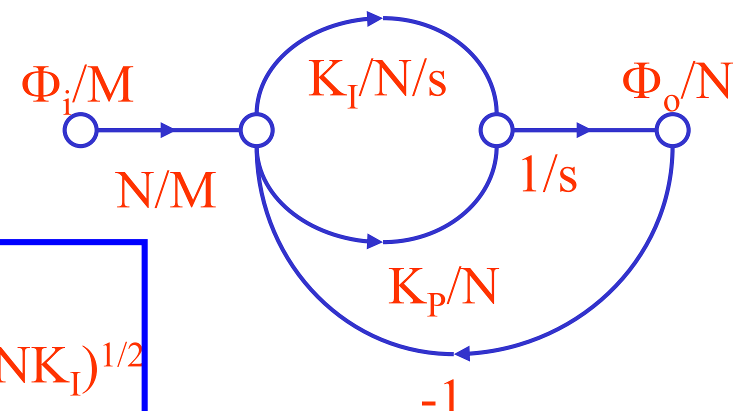
$1/Q = 2\zeta = K_P/(NK_I)^{1/2}$

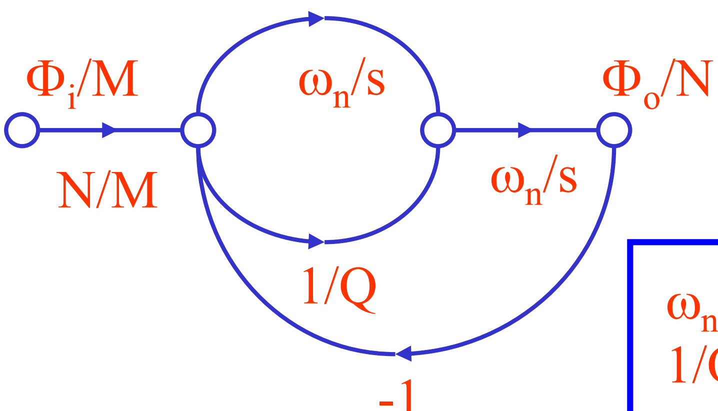
Second Order PLL Model

- Integral Gain : K_I
- Proportional Gain: K_P
- Natural Frequency
- Q-factor (Damping Factor)



$K_I = K_{PD}K_1K_{VCO}$
$K_P = K_{PD}K_2K_{VCO}$



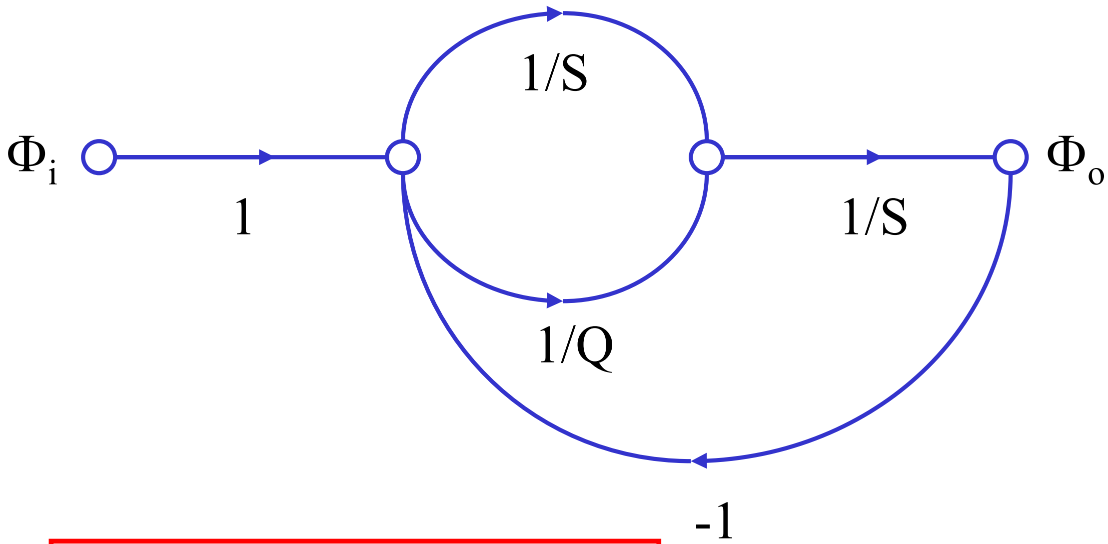


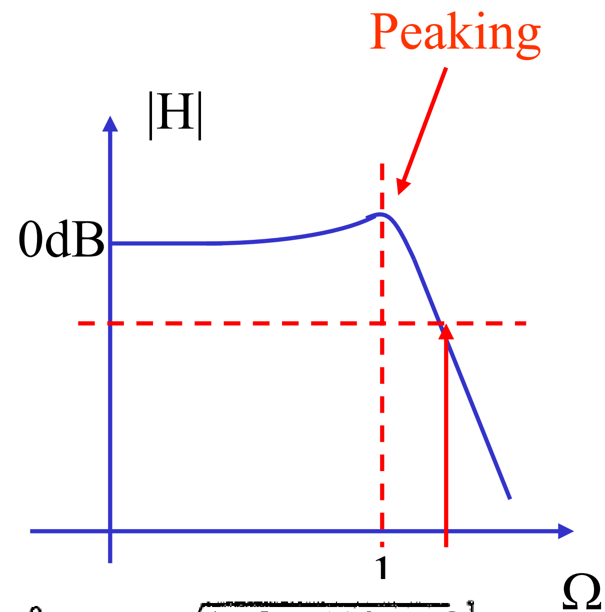
$\omega_n = (K_I/N)^{1/2}$
$1/Q = 2\zeta = K_P/(NK_I)^{1/2}$

Normalized 2^{nd} Order PLL Model

- 2^{nd} order PLL loop can be unified/normalized using two loop parameters
  - Nature frequency (ω_n)
  - Q-factor (or damping factor) ($1/Q = 2\zeta$)
  - ζ (Typically: 0.7 ~1.1)
- Actual phases can be scaled back from prototype:
  - Input phase scaling factor 1/M
  - Output phase scaling factor 1/N
  - Loop frequency scaling factor ω_n

Normalized Jitter Transfer Function





$$F_{3dB} = \frac{w_n}{2\pi}[2\zeta^2 + 1 + \sqrt{(2\zeta^2+1)^2+1}]^{\frac{1}{2}}.$$

Damping factor: $\xi = \dfrac{1}{2Q}$

$$H(s) \equiv \frac{\phi_o}{\phi_i} = \frac{\frac{1}{Q}S+1}{S^2+\frac{1}{Q}S+1}$$

Peaking effect

$$\left|H(s)\right|_{s=j\omega} = \begin{cases} 1 & \omega = 0 \\ \sqrt{1+Q^2} > 1 & \omega = 1 \\ 0 & \omega = \infty \end{cases}$$

Normalized Jitter Transfer Function

- Poles

$$S^2 + \frac{1}{Q}S + 1 = 0 \Rightarrow p_{1,2} = -\frac{1}{2Q} \pm j\sqrt{1-\left(\frac{1}{2Q}\right)^2}$$

- Or in term of damping factor

$$p_{1,2} = -\zeta \pm j\sqrt{1-\zeta^2}$$

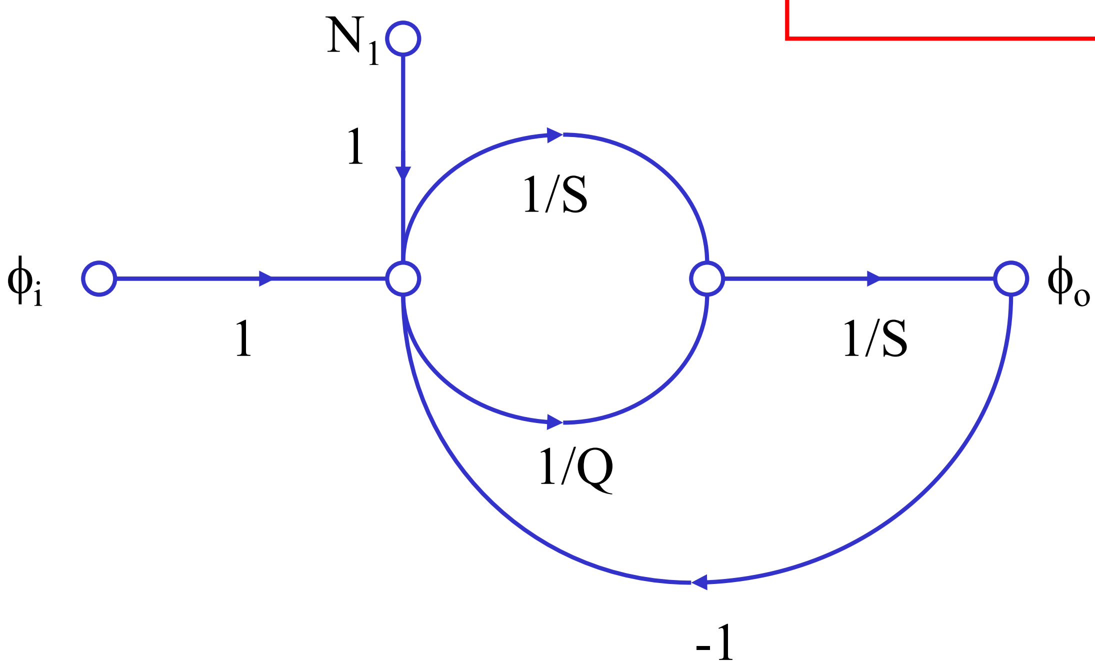
- Stability

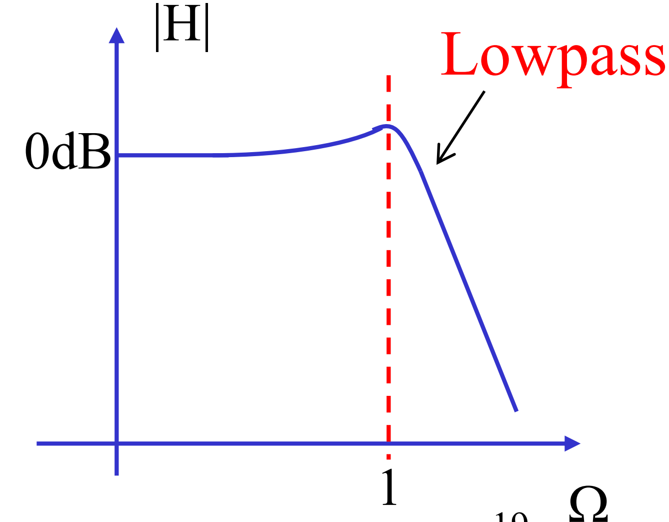
$$\zeta = \begin{cases} >1 & over\_damp \\ (0,1) & under\_damp \\ <0 & unstable \end{cases}$$

Normalized Noise Transfer Function (I)

- PFD and Charge Pump Noise injection node

$$H_{N_1}(s) \equiv \frac{\phi_o}{N_1}\Big|_{\phi_i=0} = \frac{\frac{1}{Q}S+1}{S^2+\frac{1}{Q}S+1}$$

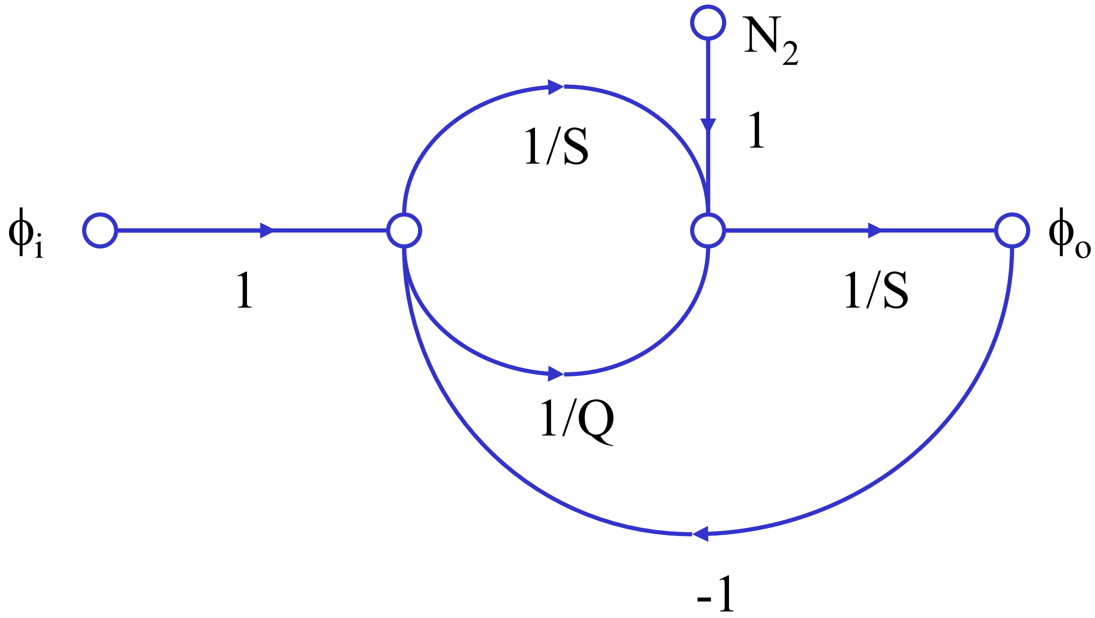


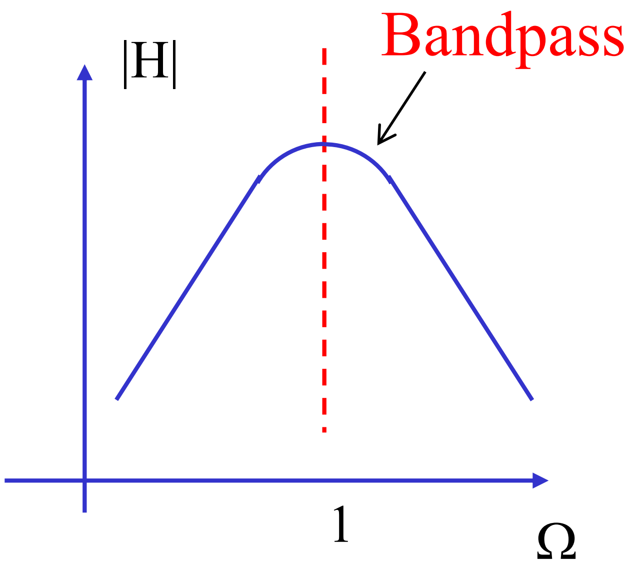


Normalized Noise Transfer Function (II)

- Loop Filter Noise Injection node

$$H_{N_2}(s) \equiv \frac{\phi_o}{N_2}\Big|_{\phi_i=0} = \frac{S}{S^2 + \frac{1}{Q}S + 1}$$

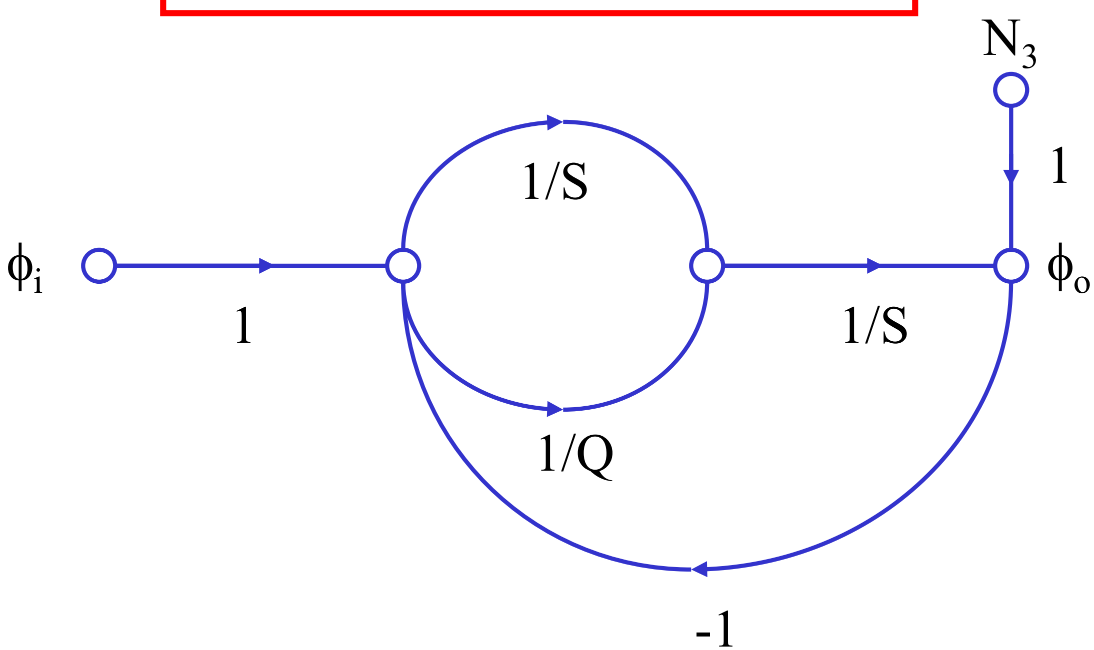


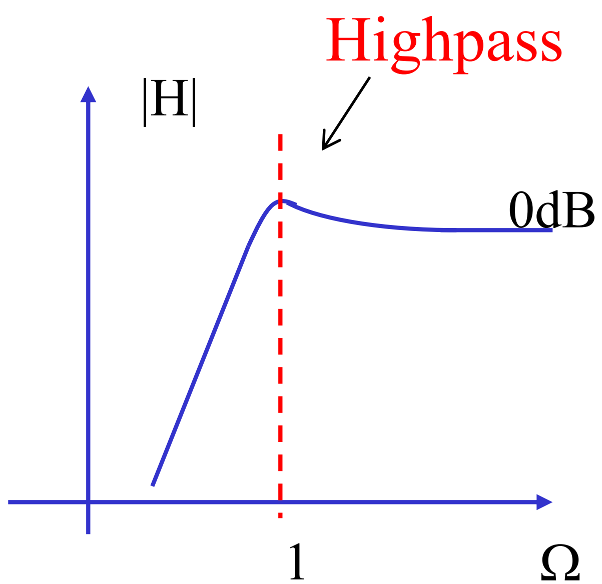


Normalized Noise Transfer Function (III)

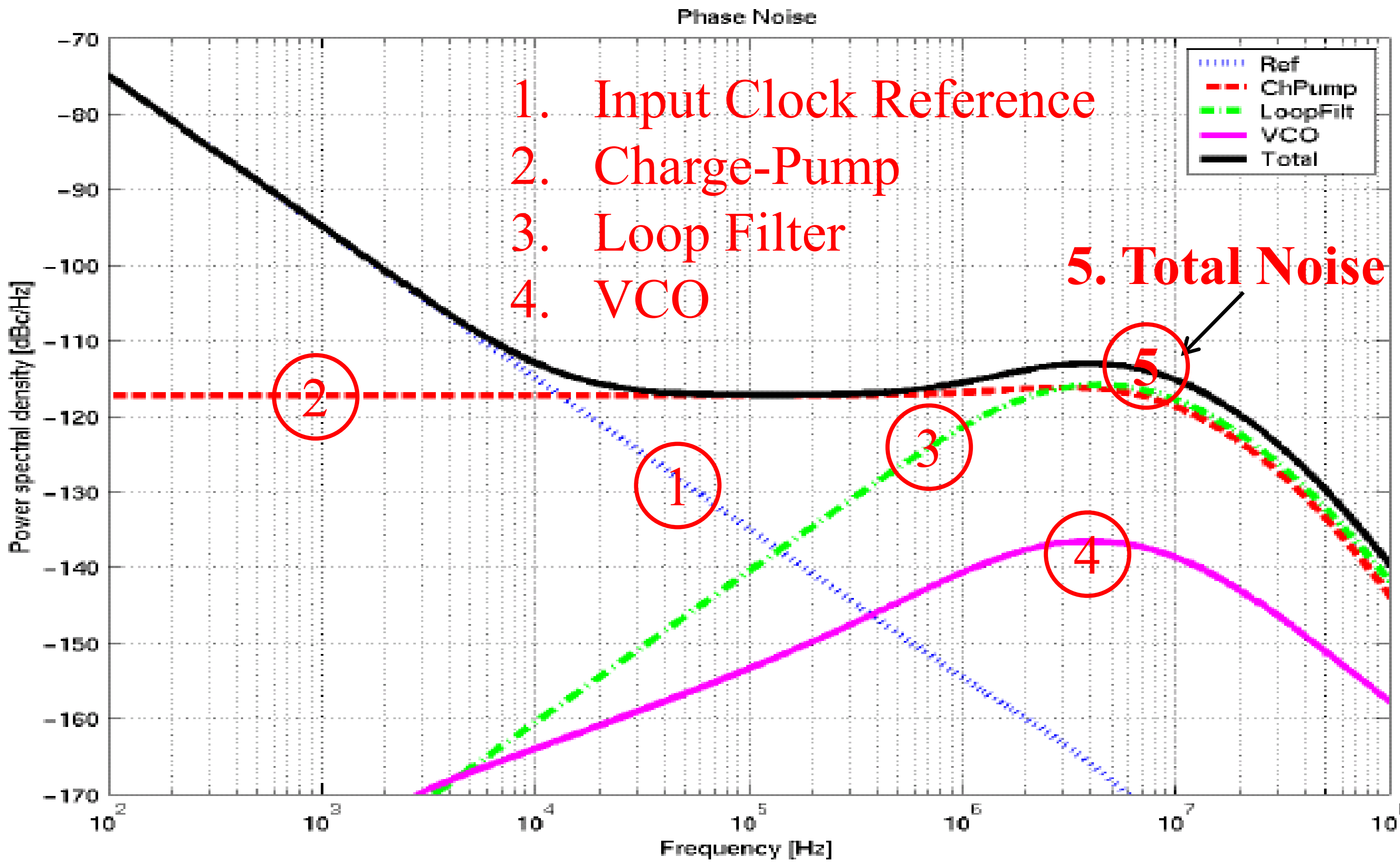
- VCO Noise Injection node

$$H_{N_3}(s) \equiv \frac{\phi_o}{N_3}\Big|_{\phi_i=0} = \frac{S^2}{S^2 + \frac{1}{Q}S + 1}$$



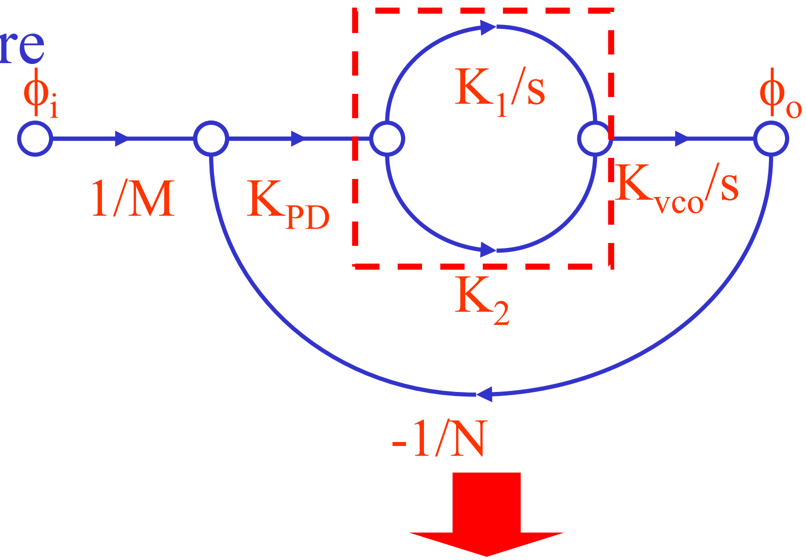


PLL Output Phase Noise (Jitter) Contributors (Noise Source + NTF)



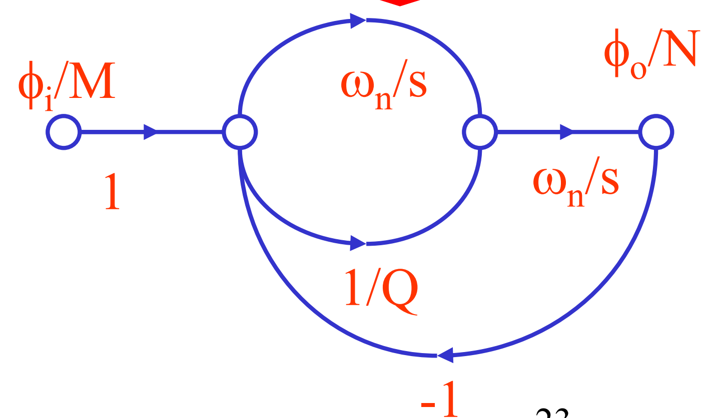
Adaptive PLL Architecture

- In conventional PLL, K_{PD}, K_1, K_2 and K_{VCO} are constants, ω_n and ζ are independent of ω_{ref}



$$\boxed{\begin{aligned}\omega_n &= (K_1 K_{PD} K_{VCO}/N)^{1/2}\\ 1/Q &= (K_2/K_1)(K_{PD}K_1K_{VCO}/N)^{1/2}\end{aligned}}$$

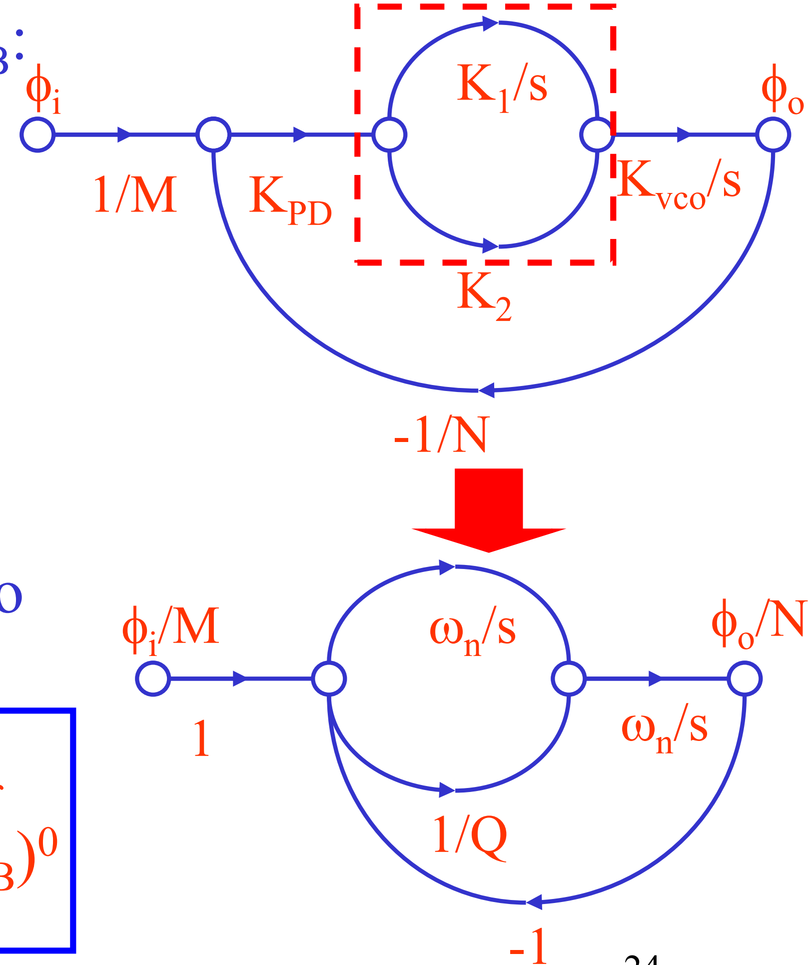
- In adaptive PLL, ω_n/ω_{ref} and Q are forced to be constant
  - No limit on operation frequency
  - Improved jitter performance
  - A ratio-based design (symmetry!)



Adaptive PLL By Self-Biasing I_B

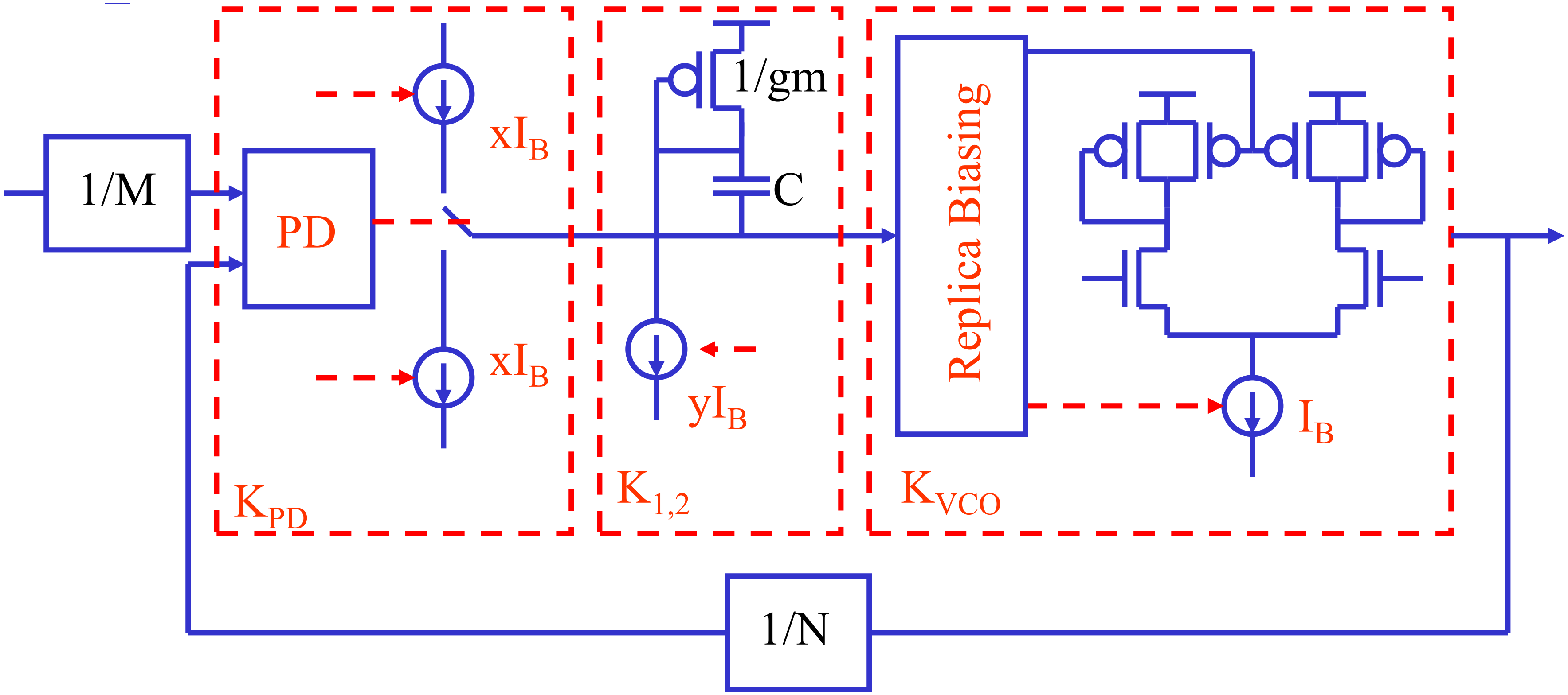
- In the self-biased PLL, if we force the parameter as function of bias I_B:
  - $\omega_{ref} \sim (I_B)^{0.5}$
  - $Kvco \sim (I_B)^0$
  - $K_{PD} \sim (I_B)^1$
  - $K_1 \sim (I_B)^0$
  - $K_2 \sim (I_B)^{-0.5}$
- This results in constant ω_n/ω_{ref} ratio and Q:

$$\boxed{\begin{aligned}&\omega_n = (K_1 K_{PD} K_{VCO}/N)^{1/2} \sim (I_B)^{0.5} \sim \omega_{ref}\\ &1/Q = (K_2/K_1)(K_{PD} K_1 K_{VCO}/N)^{1/2} \sim (I_B)^0\end{aligned}}$$



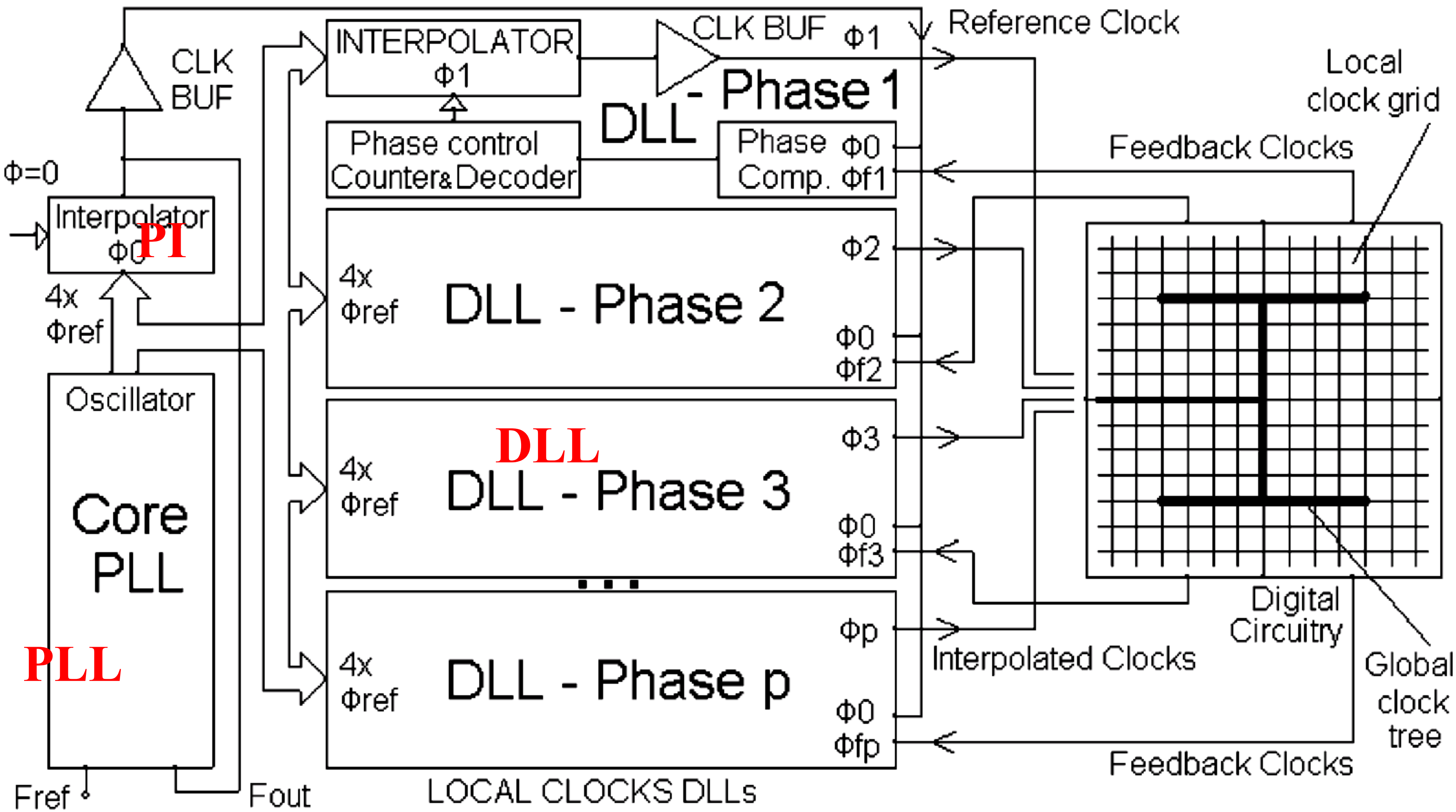
Adaptive PLL By Self-Biasing I_B

- In the self-biased PLL
  –



Note: Biasing current is not constant!!

PLL Application: CPU Clock Tree



Reading Assignment

- Please read Chapter 10 of the textbook